processes, maintenance requirements, and diagnostic capabilities. Ignoring these considerations during development often results in products that work well during demonstrations but become difficult to operate once deployed in large numbers.

Serviceability is particularly important. If technicians cannot commission a device, retrieve diagnostic logs, or resolve issues quickly during installation, maintaining thousands of deployed units becomes increasingly difficult. Designing products around field-service requirements can simplify deployment and reduce long-term maintenance costs.



Mandar Halbhavi, Deputy General Manager, Jio, speaking at India Electronics Week 2026

Five engineering lessons for large-scale deployment

Successful IoT deployments rely on much more than hardware and software. Five engineering considerations determine whether products remain reliable after deployment.

1. Prioritising customer problems over feature creep. Engineering teams often focus on adding more features, sophisticated dashboards and advanced analytics. While these capabilities may showcase technical expertise, they do not always address what customers actually need.

Most users expect a product that performs its intended function reliably, presents information clearly, and completes everyday tasks without unnecessary complexity. A stable device with a simple dashboard and meaningful reports often delivers greater value than one packed with features that are rarely used.

Effective IoT products are built around clearly defined customer requirements rather than an extensive list of features. Each function should deliver measurable value without adding unnecessary complexity.

2. Designing for real-world operating conditions. Successful IoT products are designed for the environments in which they will operate, not just the conditions in which they are developed. Factors such as temperature, dust, moisture, power quality, and network availability vary significantly across deployment sites, making field conditions far less predictable than laboratory environments.

Products intended for city-wide deployments require engineering that accounts for these variations while maintaining reliable operation. Considering real-world conditions during the design phase helps minimise failures after installation and reduces maintenance effort over the product's lifetime.

Products that operate reliably in one region may encounter very different challenges elsewhere. Temperature, humidity, dust levels, power quality, and installation conditions vary considerably between locations and must be considered during hardware design.

Products intended for city-wide deployments must be designed to withstand a wide range of environmental conditions, including regions where ambient temperatures exceed 50°C. Outdoor installations also expose devices to dust, moisture, and fluctuating power conditions that may not be encountered during laboratory testing.

Design validation should therefore extend beyond functional testing to include environmental factors likely to be experienced throughout the product's intended deployment.

3. Simplifying deployment and servicing. As deployment volumes increase, installation procedures become just as important as the product itself. Large projects cannot depend on specialist engineers visiting every site. Instead, installation should be straightforward enough for field technicians to complete quickly and consistently. Once products enter large-scale deployment, service engineers become the primary users responsible for keeping systems operational.

Large-scale IoT deployments benefit from streamlined onboarding workflows that minimise installation time and reduce the need for technical support. Users should be able to unpack a device, scan a QR code, register it with the platform, and begin receiving data without extensive configuration or technical assistance. If installation becomes complicated or requires repeated interaction with technical support, deployment times increase and customer confidence can decline.

4. Designing for product lifecycle, power, and connectivity. Reliable